METROPOLITAN GENERAL HOSPITAL
Department of Pathology and Hematology/Oncology

CLINICOPATHOLOGIC CONFERENCE CASE PRESENTATION

Date of Conference: *January 15, 2023*
Case: *Joseph Roberts*
MRN: *AML047*
DOB: *09/29/1949*
Date of Death: *December 18, 2022*
Clinical Discussant: *Dr. Sarah Williams, Hematology*
Pathology Discussant: *Dr. Robert Lee, Pathology*

CLINICAL PRESENTATION AND HISTORY:

Dr. Williams: Mr. Roberts was a 73-year-old retired postal worker who presented to his primary care physician in June 2022 with a 2-month history of progressive fatigue, unintentional 20-pound weight loss, and recurrent epistaxis. His past medical history was significant for coronary artery disease with prior CABG in 2015, type 2 diabetes mellitus, and chronic kidney disease stage 3.

Initial laboratory evaluation on 06/15/2022 revealed:

- *WBC: 2.8 K/μL with 18% circulating blasts*
- *Hemoglobin: 6.8 g/dL (baseline 11-12)*
- *Platelets: 28,000/μL*
- *Creatinine: 2.1 mg/dL (baseline 1.6-1.8)*
- *LDH: 892 U/L*
- *Uric acid: 9.2 mg/dL*

He was admitted urgently on 06/20/2022 for further evaluation. Bone marrow biopsy demonstrated:

- *Hypercellular marrow (90% cellularity)*
- *67% myeloblasts by morphology*
- *Flow cytometry: Aberrant myeloid blast population expressing CD34+, CD117+, CD13+, CD33+, HLA-DR+, CD56+*
- *Cytogenetics: Complex karyotype with 45,XY,-5,-7,+8,del(12p),der(17) and multiple additional structural abnormalities in 18/20 metaphases*
- *Molecular studies: TP53 mutation (VAF 38%), ASXL1 mutation (VAF 22%), no FLT3 or NPM1 mutations*
- *Final diagnosis: Acute myeloid leukemia with adverse-risk features (complex karyotype, TP53 mutated)*

TREATMENT COURSE:

Given his age, comorbidities, and adverse cytogenetics, intensive chemotherapy was not recommended. He was started on single-agent azacitidine 75 mg/m² days 1-7 of a 28-day cycle on 06/28/2022.

Cycle 1 (June-July 2022):

- *Complicated by tumor lysis syndrome requiring rasburicase*
- *Prolonged hospitalization for neutropenic fever*
- *Transfusion dependent (4 units pRBCs, 6 units platelets)*

Cycle 2 (August 2022):

- *Administered outpatient with dose reduction to 5 days due to cytopenias*
- *Grade 3 fatigue, spent most time in bed*

Response Assessment after Cycle 2:

- *Bone marrow: 58% blasts (stable disease)*

Cycles 3-4 (September-October 2022):

- *Further dose reductions*
- *Increasing transfusion requirements*
- *Multiple admissions for infection*

Cycle 5 (November 2022):

- *Day 8: Admitted with altered mental status, WBC 89K*
- *Evidence of leukostasis with respiratory compromise*
- *Hydroxyurea and leukapheresis performed*
- *Progressive disease declared*

TERMINAL HOSPITALIZATION:

Admitted 12/14/2022 with fever, hypotension, and respiratory distress. Initial evaluation revealed:

- *WBC 124,000 with 92% blasts*
- *Chest X-ray: Bilateral pulmonary infiltrates*
- *Blood cultures: Klebsiella pneumoniae bacteremia*

Despite broad-spectrum antibiotics, mechanical ventilation, and vasopressor support, he developed:

- *Disseminated intravascular coagulation*
- *Acute kidney injury requiring CRRT*
- *Hepatic failure*
- *Multiple organ dysfunction syndrome*

Family meeting on 12/17/2022 led to transition to comfort care. Patient expired 12/18/2022 at 14:22.

AUTOPSY FINDINGS:

Dr. Lee: Autopsy was performed 4 hours post-mortem with family consent. Key findings included:

Hematologic:

- *Bone marrow: Near 100% cellularity with sheets of myeloblasts*
- *Spleen: 680 grams (normal 150g), extensive leukemic infiltration*
- *Liver: 2,340 grams, massive leukemic infiltration with areas of necrosis*
- *Lymph nodes: Widespread involvement with myeloid sarcoma*

Cardiovascular:

- *Patent CABG grafts (LIMA to LAD, SVG to RCA and OM)*
- *Cardiomegaly (580 grams) with LVH*
- *No acute MI*

Pulmonary:

- *Bilateral pneumonia with abscess formation*
- *Diffuse alveolar damage*
- *Extensive leukemic infiltration of pulmonary parenchyma*
- *Pulmonary hemorrhage*

Central Nervous System:

- *Multiple small cerebral hemorrhages*
- *Leukemic infiltration of meninges (previously undiagnosed)*

Other:

- *Acute tubular necrosis*
- *Adrenal hemorrhage bilaterally*
- *GI tract: Multiple areas of hemorrhage*

CLINICOPATHOLOGIC CORRELATION:

The autopsy findings confirmed widespread, refractory AML with multiorgan infiltration. The complex karyotype and TP53 mutation predicted the poor response to azacitidine monotherapy. The 5-month progression from diagnosis to death (OS 6 months from diagnosis) is typical for elderly patients with TP53-mutated AML, where median survival is 4-6 months with standard therapies.

Key learning points:

1. *TP53-mutated AML remains essentially incurable with current therapies*
2. *Complex karyotype + TP53 mutation represents the worst prognostic subgroup*
3. *Even "low-intensity" therapy like azacitidine can have significant toxicity in elderly patients*
4. *CNS involvement was clinically occult but present at autopsy - consider screening in patients with very high WBC*

FINAL ANATOMIC DIAGNOSES:

1. *Acute myeloid leukemia with complex karyotype and TP53 mutation*
2. *Extensive multiorgan leukemic infiltration*
3. *Bilateral pneumonia with abscess formation*
4. *Disseminated intravascular coagulation with multiorgan hemorrhage*
5. *Multiple organ dysfunction syndrome*

Conference Moderator: Dr. James Mitchell
Transcribed by: Medical Education Office